

Infrared Small Target Detection Based on the Difference Variance Weighted Enhanced Local Contrast Measure

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ABSTRACT

Infrared Search and Tracking System (IRST) has been widely applied in many fields, but it is still challenging to detect small infrared targets in complex backgrounds. To address this problem, this paper proposes a detection framework known as Difference Variance Weighted Enhanced Local Contrast Measure (DVWELCM). First, an enhanced local contrast measure (ELCM) is used to enhance small targets and suppress complex background while improving signal clutter ratio (SCR). Second, a weighting function of the difference variance is adopted to further reduce the influence of the background and improve the robustness. Finally, by integrating enhanced local contrast measure (ELCM) and difference variance weighting (DVW), an adaptive threshold segmentation method is used to extract the real target. Extensive experiments have been performed on data sets in different scenarios. The results show that compared with the existing methods, the proposed method has better detection performance in complex backgrounds.

CCS CONCEPTS

• Computing methodologies; • Artificial intelligence; • Computer vision; • Computer vision problems; • Object detection;

KEYWORDS

Infrared search and tracking system, small target, enhanced local contrast measure, differential variance weighting function

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1 INTRODUCTION

Infrared small target detection has been widely used in many fields, such as guidance, early warning, and traffic safety systems [1]. Nevertheless, this method has certain challenges owing to long imaging distance, complex background, small area occupied by targets, and only gray information. To address these problems, many scholars have carried out studies on this technique [2].

Up to now, a large number of infrared small target detection algorithms have been proposed, including sequence based algorithm and single frame based algorithm. The former mainly use information between multiple frames to detect targets, so it requires a large amount of calculation and the detection speed is slow. By contrast, the latter only relies on one frame of image information to detect targets, so it has the advantages of small amount of calculation and high detection speed. Thus, this paper further improves the detection algorithm based on single frame.

For the past few years, the contrast mechanism of human visual system has attracted the attention of many scholars [3]. In a local range, the gray of small targets is larger, so there is a local contrast near the small target. Therefore, algorithms based on local contrast measure can achieve better performance. Such as, Kim et al.[4] captured the gray difference between the central region and its local neighborhood through LoG filter. Wang et al.[5] applied a simple DoG filter to obtain local gray difference. Chen et al.[6] employed a nine-block structure to capture the local contrast information between the target block and the surrounding background block. Han et al. [7] obtained the local gray ratio through the average gray of the target block. Qin and Li [8] pointed out that the average gray of a small block only needs to consider some maximum grays. Wei et al. [9] merged the local gray differences in two corresponding directions to suppress background edges. Han et al. [10] combined ratio operation and difference operation as local contrast measurement. Han et al. [11] put forward a local contrast algorithm based on three-layer window.

In recent years, many researchers have attempted to obtain better performance by using some joint algorithms. They proposed many algorithms that combine weighting functions with local contrast. For example, Deng et al. [12] weighted local gray difference using local entropy. Qin and Li [13] weighted the local contrast with the

improved variance; Nie et al. [14] weighted local gray difference with the standard deviation characteristic of the target. Chen and Xin [15] weighted local contrast through euclidean distance. Liu et al. [16] employed the quantity of pixel pairs between the target pixel and the corresponding surrounding background pixel. Lv et al. [17] applied regional intensity level (RIL) to calculate the statistical difference between the target and the surrounding background, and used the statistical difference to weight the local gray difference. Han et al. [3] proposed weighted strengthened local contrast measure (WSLCM) that uses a modified RIL to weight the local contrast.

In general, compared with the local contrast method, the method combining the weighting function with the local contrast measurement has better detection performance because it makes full use of the information of the target or the surrounding background. However, the current algorithms still have some shortcomings. First of all, for the design of local contrast, they only consider the enhancement of the target or the suppression of the background, but not the improvement of SCR. Second, in the design of the weighting function, the feature of pixel difference between the target and surrounding backgrounds in a certain direction is not considered as enhancement factor of the target. Third, some improved weighting functions are very sensitive to noise although they can effectively suppress background edges.

This paper proposes a detection framework known as Difference Variance Weighted Enhanced Local Contrast Measure (DVWELCM). First, inspired by the SCR calculation, the calculation of the local contrast measure (LCM) employs the standard deviation of surrounding backgrounds. Subsequently, the target and background of the raw infrared image are estimated by the nested window. After that, the ratio and difference operations are combined with the standard deviation of surrounding backgrounds to calculate ELCM. Second, the statistical discrepancy amid the target and surrounding backgrounds is seen as a weighting factor. DVW is obtained through the variance of the minimum change direction of the pixel discrepancy amid the target and surrounding backgrounds in the nested window. Finally, DVW is combined with ELCM, and then real infrared targets are detected by adaptive threshold segmentation. Experiments on some real infrared image data sets demonstrate that DVWELCM has better detectability in different backgrounds.

The rest of this paper is organized as follows. Section II introduces the proposed method. Section III presents the experimental results. Section IV concludes the paper.

2 PROPOSED ALGORITHM

The detection framework diagram of DVWELCM algorithm is shown in Figure 1. First, ELCM and DVW are calculated. The ELCM is calculated from the target estimate, background estimate and standard deviation of the surrounding background obtained by the nested window. The DVW considers the statistical discrepancy amid the target and surrounding backgrounds. True targets will be the most significant in the results obtained by ELCM and DVW. Therefore, after simple threshold segmentation, final results can easily detect the real target.

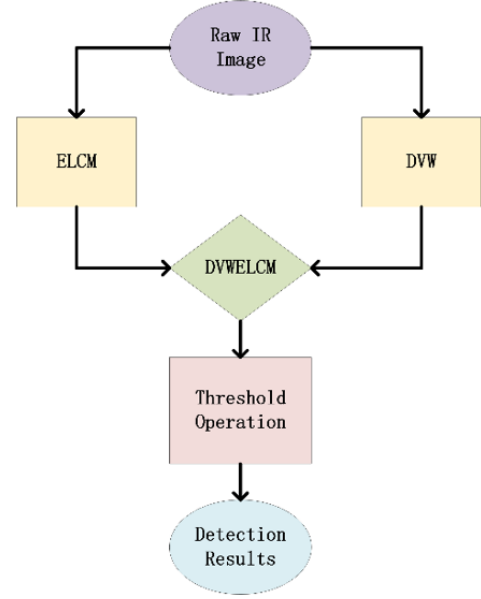


Figure 1: Flowchart of the proposed algorithm.

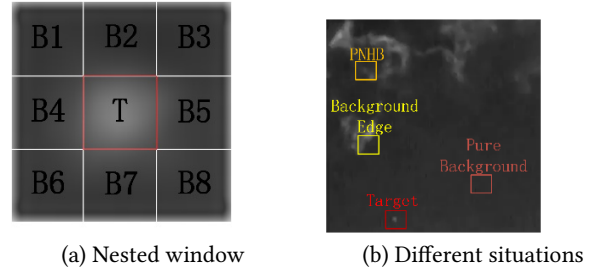


Figure 2: The information of nested window and infrared image.

2.1 Calculation of the ELCM

Inspired by Han et al. [10], this paper adopts a nested window of 9 blocks (as shown in Figure 2). Among them, block T is used to acquire the information of the target, and block Bi (i = 1, 2, ..., 8) to acquire the information of surrounding backgrounds.

The ELCM of current pixel is defined as:

$$\text{ELCM} = \min \frac{\left(H \left(\frac{M_T}{M_{Bi}} \right) (M_T - M_{Bi})^3 \right)}{\max(Std_B, \xi)}, \quad i = 1, 2, \dots, 8 \quad (1)$$

Where Bi (i = 1, 2, ..., 8) is the surrounding backgrounds blocks in eight directions, T represents the target block, M_T represents the average gray of pixels in block T whose pixel value is larger than the median gray, M_{Bi} is the mean gray of the total number of pixels in block Bi, $H \left(\frac{M_T}{M_{Bi}} \right)$ is the enhancement coefficient of the center block T of the nested window in the i-th direction, and Std_B is the average of Bi gray standard deviation of surrounding background blocks. To avoid division by 0, ξ is set to 0.0001. In addition, $H \left(\frac{M_T}{M_{Bi}} \right)$ is designed to enhance the target while suppressing the background.

$(M_T - M_{Bi})^3$ is responsible for magnifying the difference between the target and the surrounding background block while ensuring that the symbol does not change.; Std_B improves the SCR of the image.; The minimum operation between the eight directions is designed to suppress background edges. Therefore, equation 1) aims to enhance the local contrast while improving SCR. First, the average gray of the center block and the surrounding eight directions background block are calculated respectively:

$$M_T = \frac{1}{N_T} \sum_{j=1}^{N_T} G_T^j \quad (2)$$

$$M_{Bi} = \frac{1}{N_B} \sum_{j=1}^{N_B} G_{Bi}^j, \quad i = 1, 2, \dots, 8 \quad (3)$$

Where N_T is the number of pixels in T, and the pixels are larger than the median gray. N_B is the number of pixels in backgrounds Bi. G_T^j is the j-th maximum gray in center block T. G_{Bi}^j represents is the j-th maximum gray in background block Bi. Then the enhancement coefficient is:

$$H(x) = \begin{cases} x & x \geq 1 \\ 0 & x < 1 \end{cases} \quad (4)$$

Where x is the input to the function H. The standard deviation of the background block is given by:

$$Std_{Bi} = \sqrt{\frac{1}{N_B} \sum_{j=1}^{N_B} [G_{Bi}^j - M_{Bi}]^2} \quad i = 1, 2, \dots, 8 \quad (5)$$

$$Std_B = \frac{1}{8} \sum_{i=1}^8 Std_{Bi} \quad (6)$$

where Std_{Bi} is the gray standard deviation of background block Bi. Through the above calculation, the ELCM of each pixel in image can be obtained.

2.2 Calculation of the DVW

With reference to Gao et al. [18], the weighting function is the variance of the difference between the target and surrounding backgrounds in a certain direction. Then, DVW of a pixel (x, y) is calculated as follows:

$$DVW_i(x, y) = Var(T - Bi), \quad i = 1, 2, \dots, 8 \quad (7)$$

$$DVW(x, y) = \min(DVW_i(x, y)), \quad i = 1, 2, \dots, 8 \quad (8)$$

Where $(T-Bi)$ denotes that the target block T and the surrounding background block Bi carry out the corresponding pixel difference operation. $Var(T-Bi)$ is used to calculate the variance of the input $(T-Bi)$. The nine blocks have a same size of $n \times n$.

2.3 Calculation of the DVWELCM

In actual application, it is very important to perform multiscale operations because the size of the target is often unpredictable. Firstly, different values are assigned to the block size, so as to calculate ELCM and DVW for each pixel (x,y). Subsequently, multiscale

DVWELCM can be acquired through the maximum pooling operation on ELCM and the minimum pooling operation on DVW:

$$DVWELCM(x, y) = \max\{ELCM_p(x, y)\} \times \min\{DVW_p(x, y)\}, \quad p = 1, 2, \dots, s \quad (9)$$

Where p is the p-th scale, and s is the number of scales

2.4 NORMALIZATION

Normalization is required to adjust the image to a specific gray range. For the raw infrared images processed by DVWELCM algorithm (denoted by IDVWELCM), its maximum gray is $\max(IDVWELCM)$ and minimum gray is $\min(IDVWELCM)$. Its calculation formula is:

$$IDVWELCM(x, y) = (2^n - 1) \frac{IDVWELCM(x, y) - \min(IDVWELCM)}{\max(IDVWELCM) - \min(IDVWELCM)} \quad (10)$$

Where n is the factor that adjusts the gray range. In this paper, we take 8, that is, the grayscale range is 0-255.

2.5 TARGET SEGMENTATION

After the saliency map $IDVWELCM$, adaptive threshold segmentation is adopted to detect the real target from backgrounds. Its calculation formula is:

$$Th = \alpha \max(IDVWELCM) + (1 - \alpha) \text{mean}(IDVWELCM) \quad (11)$$

Where $\text{mean}(IDVWELCM)$ is the average of $IDVWELCM$, α is a factor between 0 and 1. Our experiments show that a α between 0.4 and 0.8 is suitable for single target detection. The calculation of DVWELCM, normalization process and threshold segmentation are combined and called the DVWELCM algorithm.

2.6 DVWELCM DISCUSSION

Before the theoretical analysis, it should be pointed out that DVWELCM is very suitable for parallel computing. Therefore, it is very suitable for multi-core computing scenarios. A detailed description of DVWELCM shows the effectiveness of the method. Next, this paper will discuss the results in different situations (Figure 2 b).

For the target pixel in the center block T, the target is usually locally prominent. This means M_T is large, M_{Bi} is small, and Std_B is small, so its ELCM and DVW are large. Therefore, the value of the resulting DVWELCM will be large.

For a pixel in a pure background, the gray of backgrounds is often homogeneous in a larger range. This means that M_T is close to M_{Bi} and Std_B to 0, so its ELCM is close to 0. And its DVW is approximately zero. Therefore, the final DVWELCM value is close to 0.

Therefore, M_T may be greater than M_{Bi} , but Std_B is large, so its ELCM is small. Moreover, a region around it may contain the same edge as the central region, thus, its DVW is quite small, and the background edge can be further suppressed.

For a (pixel-sized noises with high brightness) PNHB [8], PNHB usually appears as a single pixel. Since the pixel gray used in the definition of M_T is greater than the median pixel gray, rather than the single maximum pixel gray, its M_T is smaller than the real target,

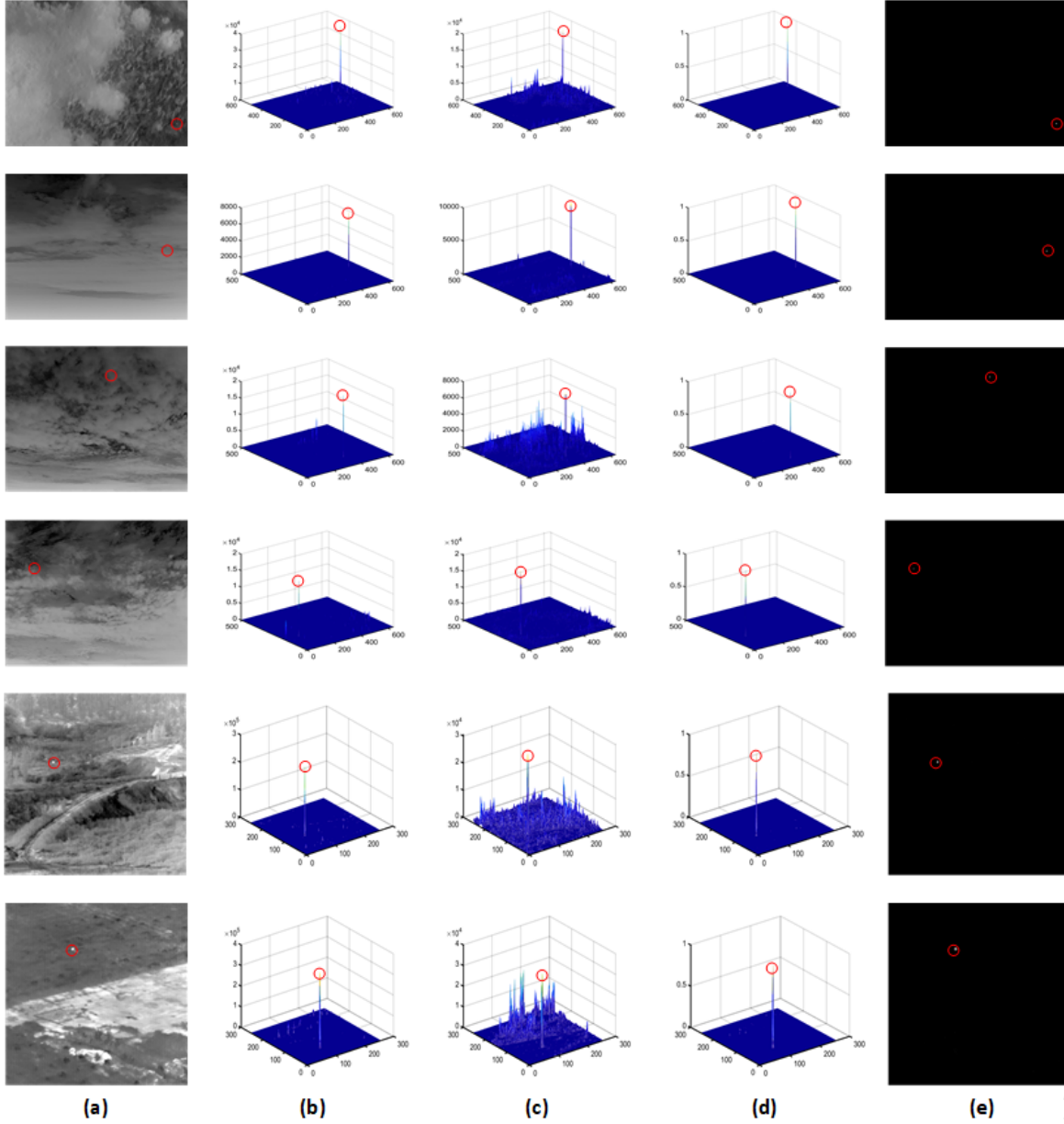


Figure 3: From top to bottom: Group A-F. (a) Raw images; (b) ELCM; (c) DVW; (d) DVWELCM (normalized); (e) Final detection results

so its ELCM will be smaller than the real target. Moreover, its DVW is small, so PNHB can be further inhibited.

3 EXPERIMENTAL RESULTS

The simulations and experiments are carried out on a computer with an 8GB memory and a 3.0GHz Intel i7-9700 processor. The code is implemented in MATLAB R2018a.

In order to estimate the detection capability of different algorithms, experiments are performed on real image data sets (represented by Group A-F). Table 1 shows the details of the tested data.

First, the size of a block at each scale is determined. In the experiment, four scales are adopted, and each block is determined to be 3×3 , 5×5 , 7×7 and 9×9 respectively. Figure 3 displays the resulting diagrams for every phase. Figure 3(a) is the raw images. For every dataset, a sample is selected and all targets are marked circles. It can be observed from the figure that the target is often submerged in

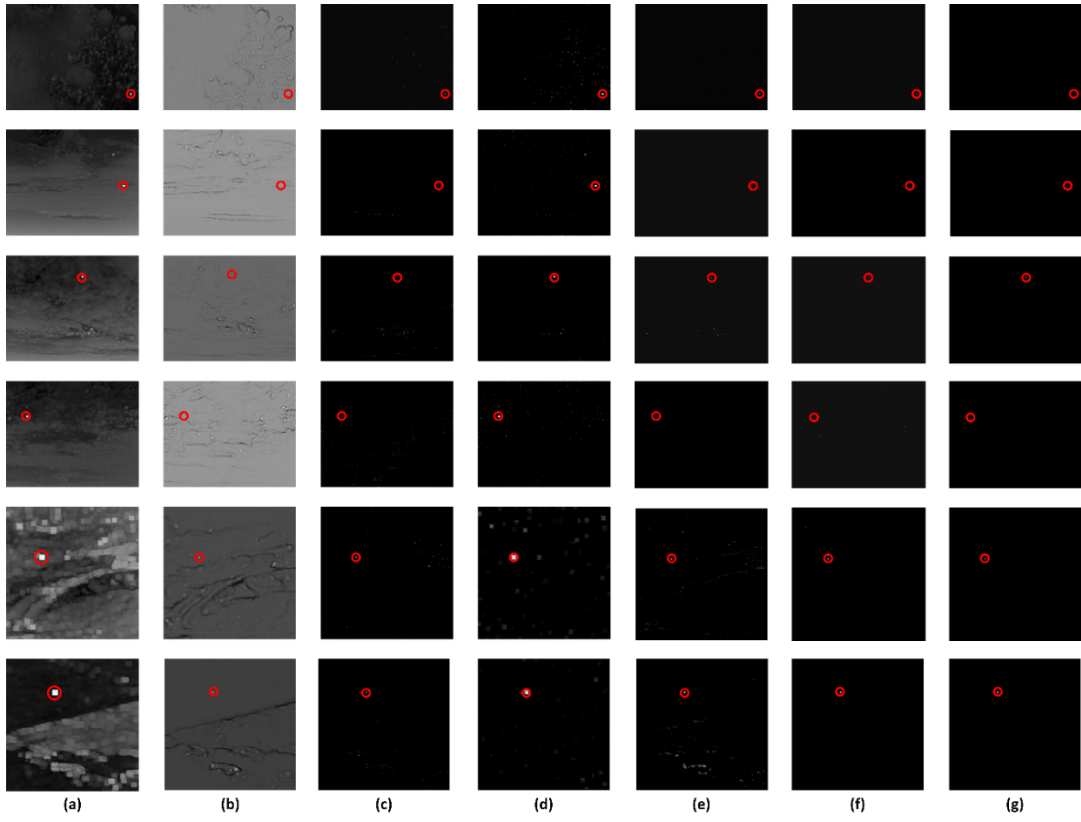


Figure 4: Detection results. (a) LCM, (b) MPCM, (c) HBMLCM, (d) RLCM, (e) LIG, (f) WSLCM, (g) Proposed Method

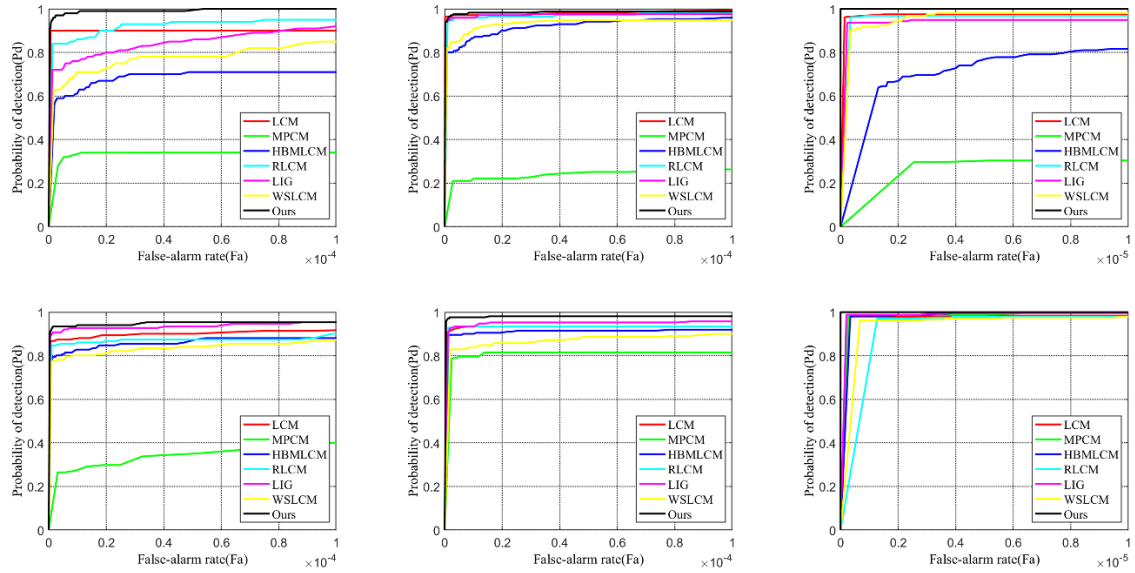


Figure 5: The ROC curves of Group A-F

Table 1: Details of the tested data

	Frame	Target Type	Background Information
Group A	100	Plane target	gloomy sky with heavy cloudy clutters
Group B	171	Plane target	a comparatively smooth sky background
Group C	158	Plane target	Complex sky background
Group D	151	Plane target	heavy cloudy clutter
Group E	211	Helicopter target	Complex land background
Group F	243	Helicopter target	Complex land background

Table 2: Different algorithms for SCRG and BSF

Group	LCM	MPCM	HBMLCM	RLCM	LIG	WSLCM	Ours
A	24.2752	13.2793	30.8718	32.5553	34.6260	34.1364	50.1494
SCRG							
B	22.2550	14.0645	46.8187	34.9276	51.0748	53.9550	65.7052
C	24.1474	20.3064	49.7478	37.7600	58.5937	66.6703	86.6807
D	22.9577	9.3852	32.1463	33.75552	35.0325	37.8317	64.2370
E	13.8816	14.0444	29.9637	23.9322	29.7245	34.8566	42.3721
F	6.9815	15.5991	26.3154	16.5679	24.4274	31.1022	34.9384
A	2.4822	5.2802	15.8665	7.4064	41.5914	75.1530	275.2556
BSF							
B	1.7173	8.8523	138.1782	9.9105	1.7600E3	1.3673E4	3.2739E4
C	2.0318	13.4154	142.1896	10.3252	6.2113E3	7.4325E4	3.4656E5
D	2.4307	6.1093	27.8160	9.4574	191.6220	1.2081E3	1.0006E4
E	1.5176	6.1773	23.1909	4.0988	20.3292	68.8771	115.8675
F	0.8889	6.3982	16.7103	3.2660	9.4410	35.4064	42.6299

complex backgrounds, which poses a great challenge for detection. The results of ELCM are shown in Figure 3(b). It can be seen that the ELCM of the target is the most prominent and the ELCM of the background is extremely suppressed. However, there may still be some background residuals for complex backgrounds. Figure 3(c) shows the result of DVW. It can be observed that the DVW of the target is large and the DVW of the complex background is small. Figure 3(d) shows the result of multiscale DVWELCM. It can be seen that the DVWELCM of the target is the largest and the DVWELCM of the complex background is quite small, close to zero. Figure 3(e) is the final results.

Subsequently, to verify the detection capability of DVWELCM, six most advanced algorithms are selected for comparison: Chen's LCM algorithm [6], Wei's MPCM algorithm [9], Shi's HBMLCM algorithm [19], Zhang's LIG algorithm [20], Han's RLCM algorithm [10] and WSLCM algorithm [3]. The raw infrared image employed in Figure 3 is also used in Figure 4. The processing results of each algorithms are shown in Figure 4. It can be observed from the figure that although LCM and MPCM can successfully detect targets, there are a bunch of false alarm points. HBMLCM, LIG and RLCM can not only detect targets, but also significantly reduce the number of

false alarm points in the image compared with the detection results of LCM and MPCM. The detection effect of WSLCM is better, and only when the background of the image is complex, a few false alarm points will appear in the detection result. By contrast, the proposed method can detect the target without any false alarm points and well suppress the clutter and noise at the same time.

In order to better reflect the superiorities of DVWELCM, SCR gain (SCRG) and background suppression (BSF) [17] are employed to estimate the detection capability, as shown in Table 2.

They are defined as:

$$\text{SCRG} = 20\log_{10} \left(\frac{\text{SCR}_{out}}{\text{SCR}_{in}} \right) \quad (12)$$

$$\text{BSF} = \frac{\sigma_{in}}{\sigma_{out}} \quad (13)$$

Where SCR_{in} and SCR_{out} are SCR of the raw image and ID-VWNLCM respectively. σ_{in} and σ_{out} are the standard deviations of the raw image and IDVWELCM respectively. It can be observed from the table that the proposed method is superior to other algorithms in SCRG and BSF.

Finally, false alarm rate (Fa) and probability of detection (Pd) [18] are used to draw the ROC curve shown in Figure 5. It can be

observed from the figure that, in most cases, the proposed method has good detection performance in different background and clutter. They are defined as:

$$Pd = \frac{\text{number of total detection}}{\text{number of real targets}} \quad (14)$$

$$Fa = \frac{\text{number of false pixels detection}}{\text{number of total pixels in images}} \quad (15)$$

4 CONCLUSIONS

To address the problem of infrared small target detection with complex background, DVWELCM detection framework is proposed in this paper. First, the ELCM is calculated from the target estimate, background estimate and standard deviation of the surrounding background obtained by the nested window. Then, the weighting function DVW takes into account the statistical discrepancy amid the target and surrounding backgrounds. Finally, combined with ELCM and DVW, adaptive threshold segmentation is employed to extract real objects. Experimental results demonstrate that DVWELCM can acquire higher Pd and lower Pd, thus it has better robustness.

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